Milestone-3: Geographical Pattern Visualization

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1. Project Description

The Visualizing US Natural Disaster Declarations project is focused on analysing historical disaster data to uncover patterns, trends, and insights that can support better understanding of disaster occurrences across the United States. While earlier milestones emphasized temporal trends, this milestone shifts the focus toward geographical analysis, which plays a crucial role in identifying region-specific disaster patterns.

The primary objective of Milestone-3 is to analyse how disaster declarations are distributed across different states and regions. By leveraging geospatial visualization techniques such as choropleth maps, bar charts, and stacked visualizations, this milestone aims to identify high-risk areas, understand regional variations in disaster types, and explore the relationship between geography and disaster frequency.

Geographical analysis helps in identifying disaster-prone regions (hotspots), understanding environmental influences, and supporting effective disaster management strategies. The insights derived from this analysis can assist policymakers, disaster response teams, and researchers in improving preparedness, mitigation, and resource allocation.

2. Dataset Used

The dataset used in this milestone is the FEMA Disaster Declarations dataset, which contains detailed records of federally declared disasters across the United States. This dataset serves as a comprehensive source for analysing disaster trends over time and across different regions.

Each entry in the dataset represents a disaster declaration and includes several key attributes:

- State – The US state where the disaster occurred
- Incident Type – Category of disaster (e.g., Flood, Storm, Fire, Hurricane)
- Declaration Date – Date when the disaster was officially declared
- Incident Begin Date – Start date of the disaster event
- Incident End Date – End date of the disaster
- Region – FEMA administrative region
- Fiscal Year Declared – Year of declaration

The dataset consists of thousands of records spanning multiple decades, providing sufficient data to perform meaningful geographical analysis. Its wide coverage enables the identification of long-term regional patterns and variations in disaster occurrences.

3. Environment Setup

The analysis was conducted using Python within a Jupyter Notebook environment, which allows for seamless integration of code, visualizations, and explanatory text.

The following libraries were used:

- **Pandas**
  Used for loading, cleaning, and manipulating structured data. It provides powerful data structures such as Data Frames, enabling efficient grouping and aggregation operations.

- **NumPy**
  Used for performing numerical computations and handling array-based data efficiently.

- **Matplotlib**
  A core visualization library used to create static graphs such as line charts and bar charts.

- **Seaborn**
  Built on top of Matplotlib, Seaborn provides enhanced visualization capabilities with improved aesthetics and statistical plotting functions.

- **Plotly Express**
  Used for creating interactive visualizations, particularly choropleth maps, which are essential for representing geographical data across US states.

The combination of these tools enables comprehensive data analysis and the creation of clear, visually appealing graphs for interpretation.

4. Data Exploration

Before performing geographical analysis, it is essential to explore the dataset to understand its structure, quality, and potential issues.

The following steps were carried out:

4.1 Dataset Structure Verification

The dataset was loaded into a Pandas Data Frame, and its structure was examined to understand the number of rows, columns, and available features.

4.2 Column Inspection

Column names and their respective data types were reviewed to ensure that relevant fields such as state and incident Type were correctly formatted.

4.3 Unique State Values

The unique values in the state column were inspected to verify consistency. This step ensured that state codes were standardized and suitable for mapping.

4.4 Missing Value Analysis

Missing values were checked across key columns. Ensuring completeness of the state column is especially important for geographical analysis.

4.5 Distribution Overview

The distribution of disaster declarations across different states and incident types was briefly analysed to gain initial insights.

This exploration phase ensured that the dataset was reliable and ready for further preprocessing and analysis.

5. Data Preprocessing

Data preprocessing is a critical step that prepares raw data for analysis by ensuring accuracy and consistency.

The following preprocessing steps were performed:

5.1 Handling Missing Values

Missing or null values in important columns such as state were identified and handled appropriately. Records with missing state information were either removed or corrected.

5.2 Data Standardization

State codes were standardized to ensure consistency across the dataset. This step is crucial for correctly mapping states in geographical visualizations.

5.3 Aggregation of Disaster Counts

The dataset was grouped by state to calculate the total number of disaster declarations per state. This aggregation forms the basis for geographical comparison.

5.4 Multi-Dimensional Aggregation

To analyse disaster types geographically, the dataset was grouped by both state and incident Type. This allows for deeper insights into how different disaster categories vary across regions.

5.5 Data Transformation

The aggregated data was reshaped into suitable formats (e.g., pivot tables) for visualization purposes.

These preprocessing steps ensured that the data was clean, structured, and ready for accurate geographical analysis.

6. Geographical distribution of disasters

To understand how disasters are distributed across the United States, the total number of disaster declarations was calculated for each state.

A bar chart visualization was created to highlight the top states with the highest number of disasters.

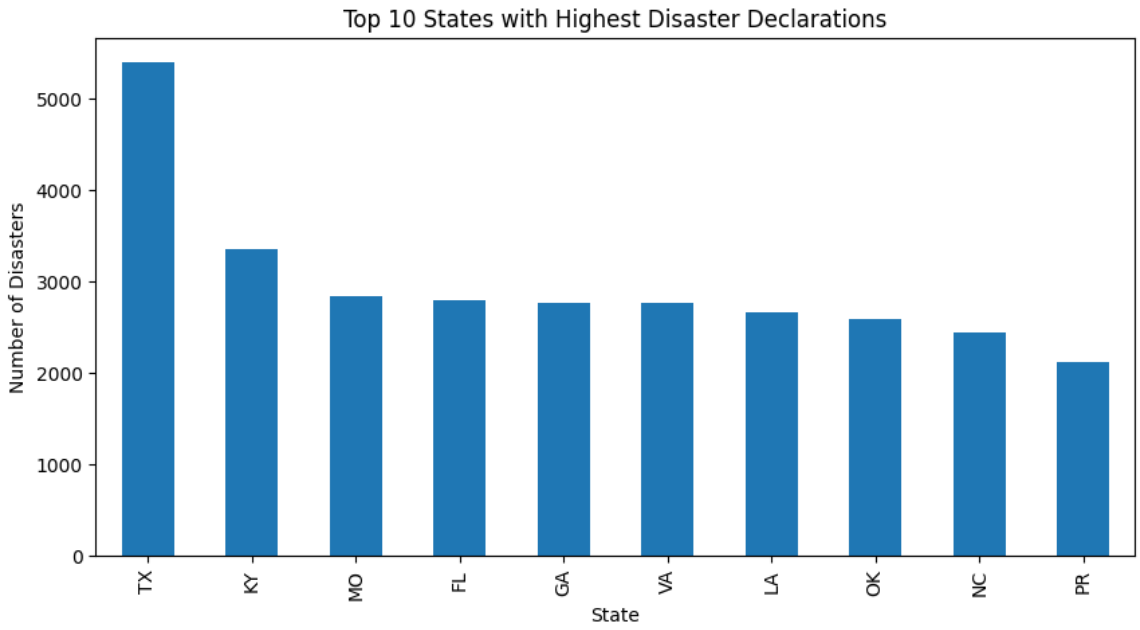
Observation: The chart clearly shows that certain states experience a significantly higher number of disaster declarations compared to others.

Pattern: States such as Texas, California, and Florida consistently rank among the top disaster-prone states.

Interpretation: These states are geographically more vulnerable due to factors such as:

- Coastal exposure to hurricanes
- Wildfire-prone environments
- High population density and infrastructure

This analysis provides a clear understanding of which states are most affected by disasters.



7. Choropleth Map Visualization

A **choropleth map** was used to visualize disaster distribution across US states. In this map, each state is color-coded based on the number of disaster declarations.

Observation:

The map visually highlights regions with high and low disaster frequencies.

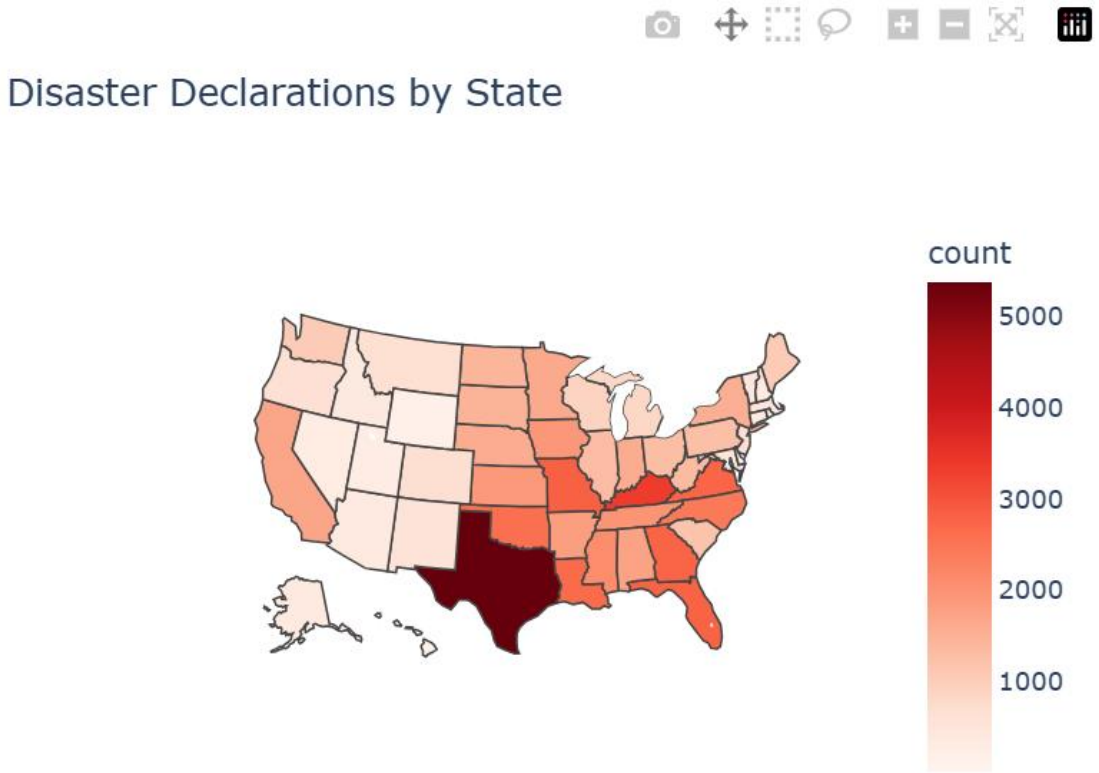
Pattern:

- Southern and coastal states show darker shades, indicating higher disaster counts
- Central and northern states show relatively lower intensity

Interpretation:

The geographical location of states plays a major role in disaster occurrence. Coastal states are more exposed to hurricanes and storms, while western states face risks such as wildfires.

The choropleth map provides an intuitive and visually effective way to understand spatial patterns in disaster data.



8. Incident Type Analysis by State

To further explore geographical patterns, disaster data was analyzed based on both state and incident type.

A stacked bar chart was used to visualize the contribution of different disaster types in each state.

Observation:

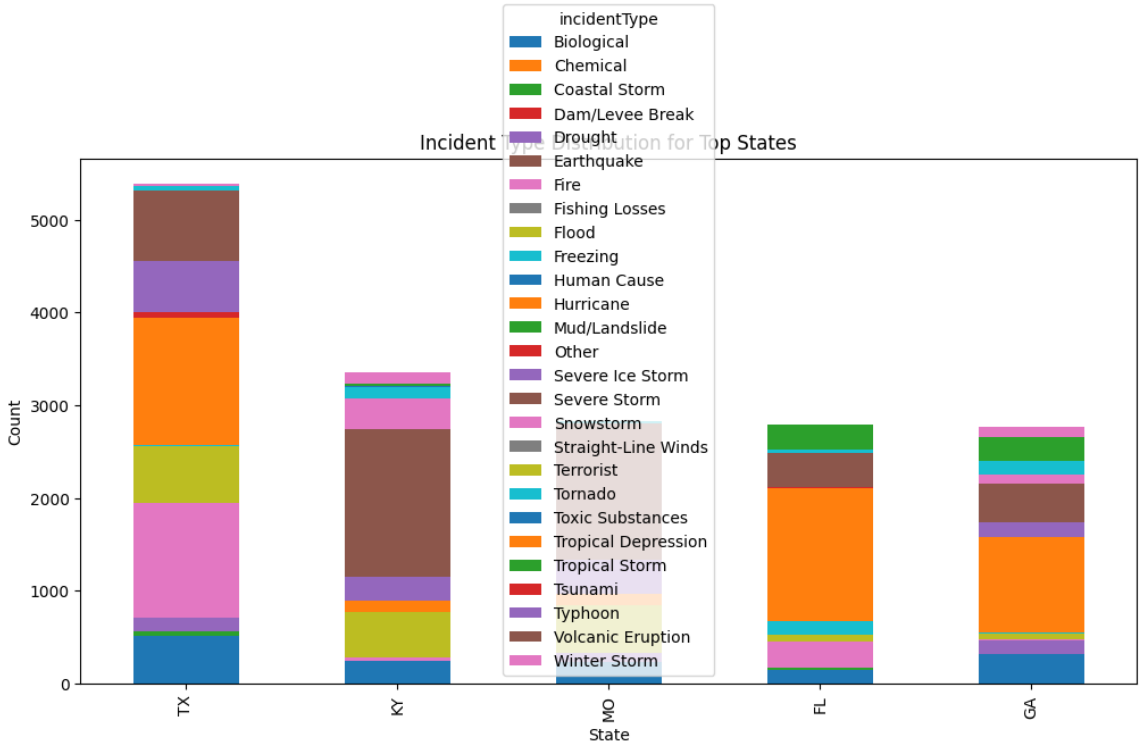
Different states exhibit different dominant disaster types.

Pattern:

- Storms are the most common disaster type across most states
- Wildfires are more prominent in states like California
- Floods are common in regions with major river systems

Interpretation:

The type of disaster is strongly influenced by environmental and climatic conditions specific to each region. This analysis helps in understanding the nature of disasters affecting different states.



9. Hotspot Analysis

Hotspot analysis was conducted to identify regions that are particularly prone to specific types of disasters.

For example, a choropleth map was created to visualize the distribution of **flood-related disasters**.

Observation:

Certain states show significantly higher occurrences of specific disaster types.

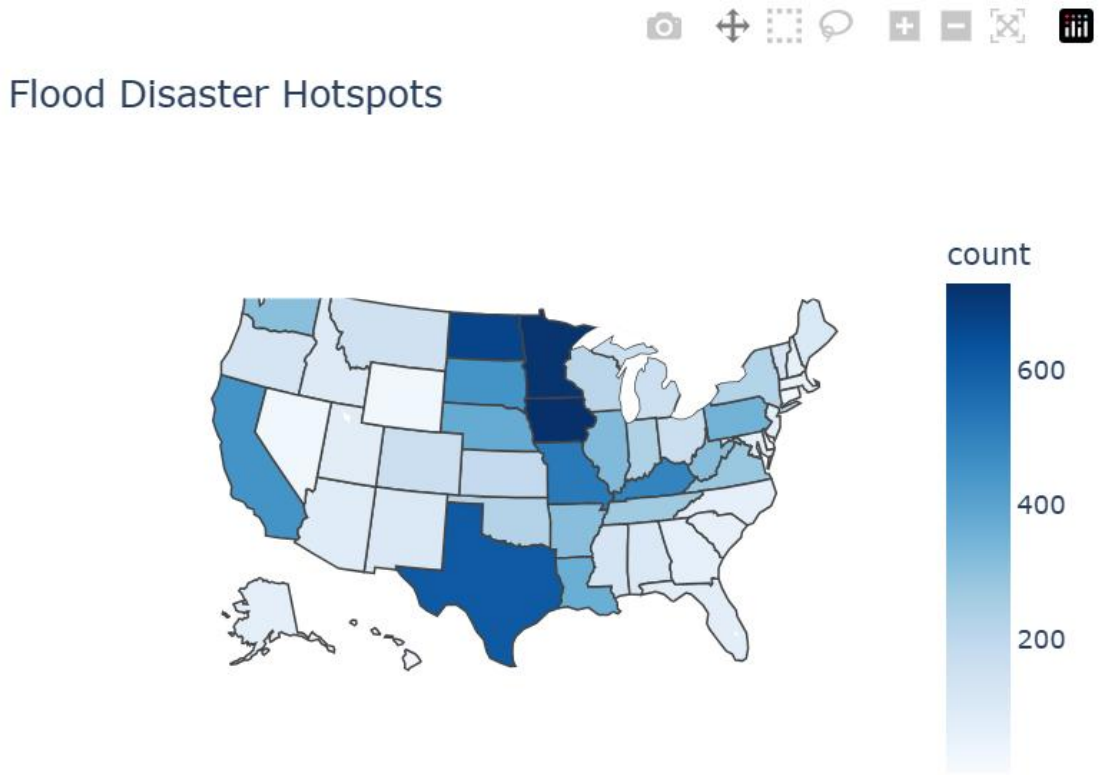
Pattern:

Flood and hurricane hotspots are concentrated in coastal and southern regions.

Interpretation:

Geographical factors such as proximity to water bodies, climate conditions, and terrain contribute to higher disaster risks in these regions.

Hotspot analysis provides targeted insights that are useful for disaster mitigation planning.



10. Conclusion

The geographical analysis of FEMA disaster declarations provides valuable insights into the spatial distribution of disasters across the United States.

Key findings include:

- Disaster occurrences are unevenly distributed across states
- Coastal and southern states are more disaster-prone
- Certain states consistently experience higher disaster frequencies
- Different regions are affected by different types of disasters
- Environmental and geographical factors significantly influence disaster patterns

This milestone demonstrates the importance of geographical analysis in understanding disaster trends. The insights obtained can support better disaster preparedness, efficient resource allocation, and informed decision-making for disaster management.